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(54) **GRIPPER FOR MANIPULATING A DEVICE FOR IDENTIFYING AN ANIMAL AND/OR REMOVING TISSUE FROM AN ANIMAL COMPRISING HOLDING MEANS WITH REMOTE ACTUATION**

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See application file for complete search history.

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(57) **ABSTRACT**

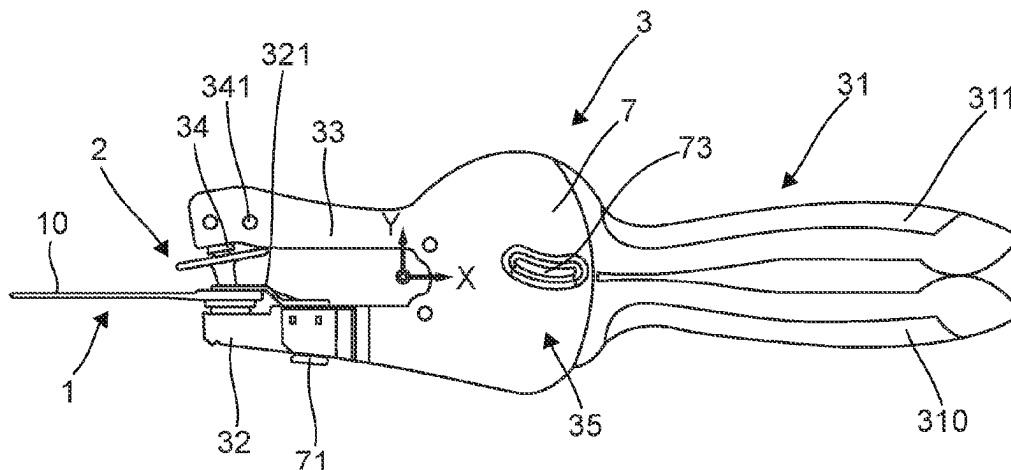
The present invention relates to a clamp (3) for handling a device for identifying an animal and/or sampling animal tissue, said clamp (3) comprising two jaws (32, 33), a handle (31) for moving said jaws (32, 33) together, and means (321) for securing an element of said identification and/or sampling device to one of said jaws (32, 33), said securing means (321) being movable between:

a release position, and

a securing position.

According to the invention, said clamp (3) comprises actuation means (73) for moving said securing means (321) from

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one to the other position thereof, said actuation means (73) being actuatable via said handle (31).

18 Claims, 6 Drawing Sheets

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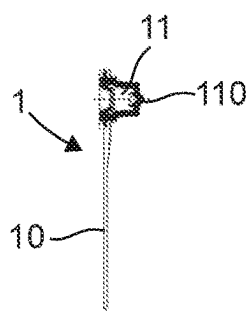


Fig. 1b

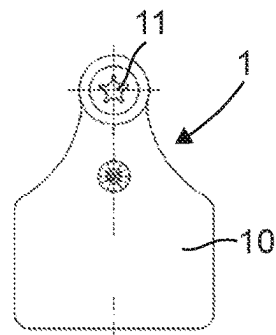


Fig. 1a

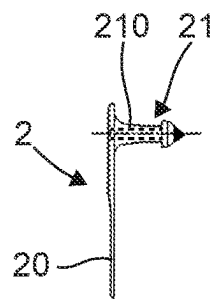


Fig. 2b

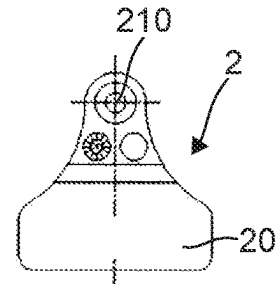


Fig. 2a

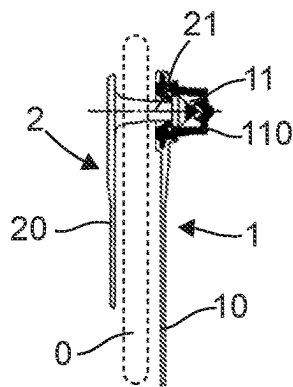


Fig. 4

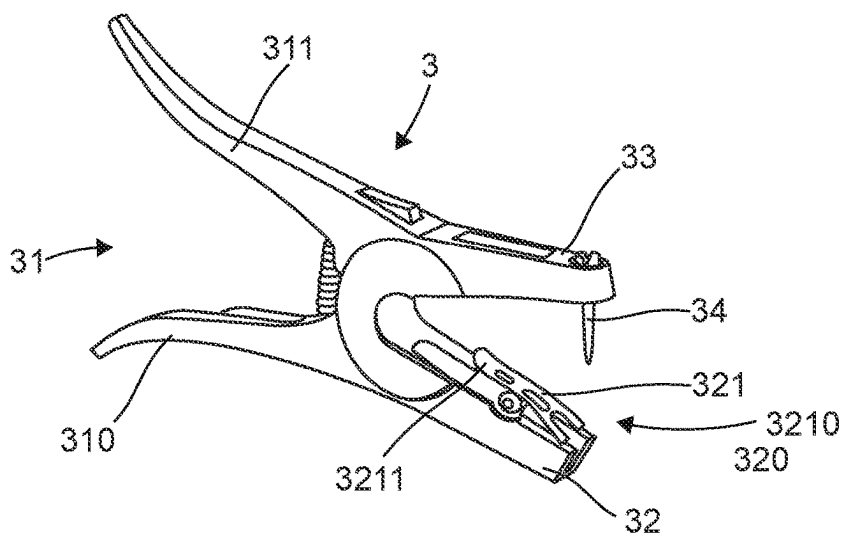


Fig. 3

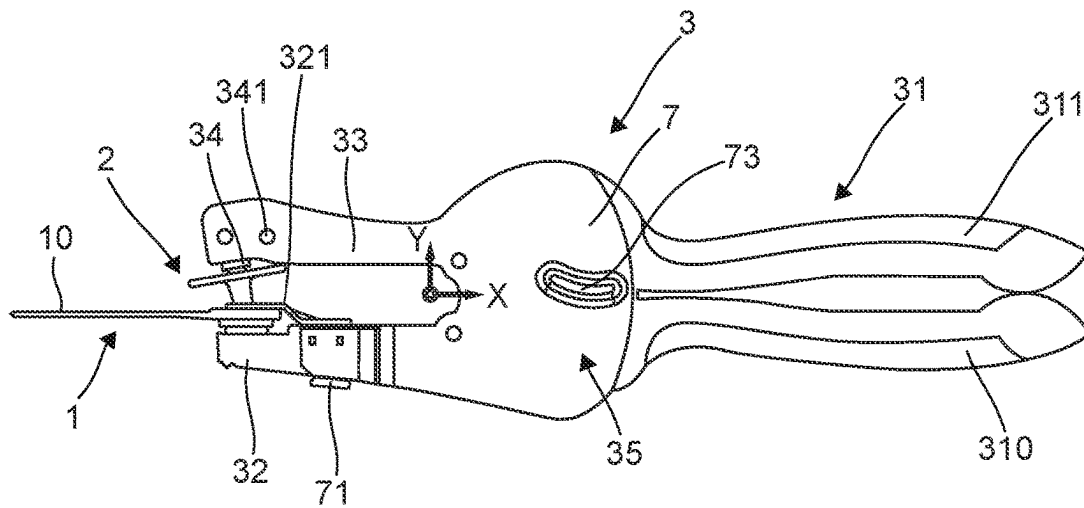


Fig. 5

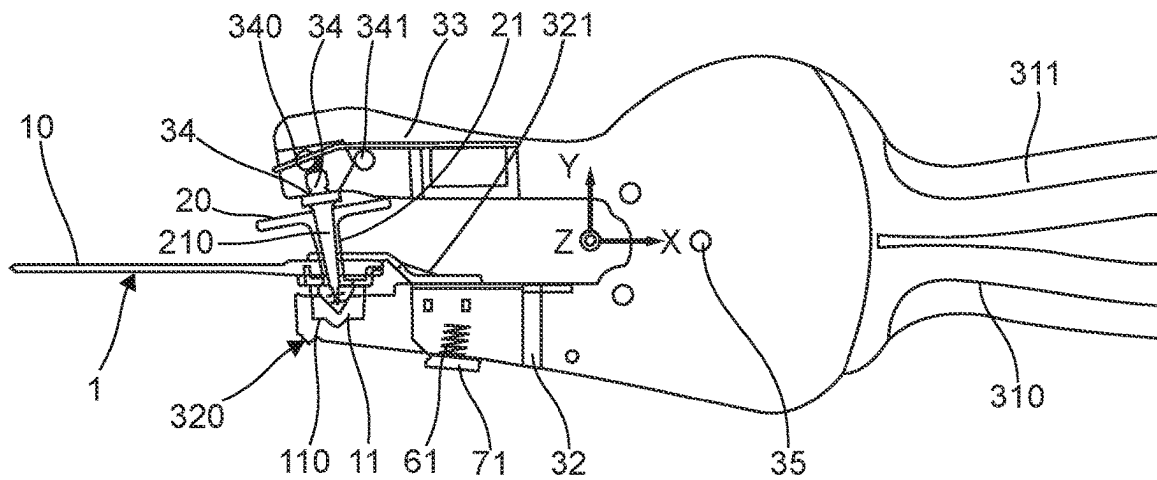


Fig. 6

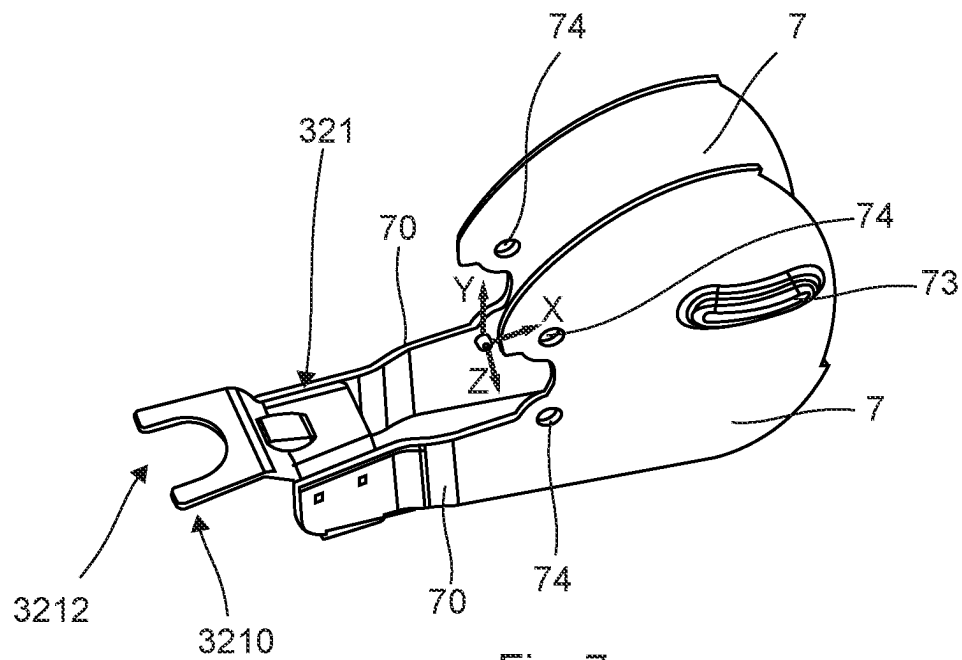


Fig. 7

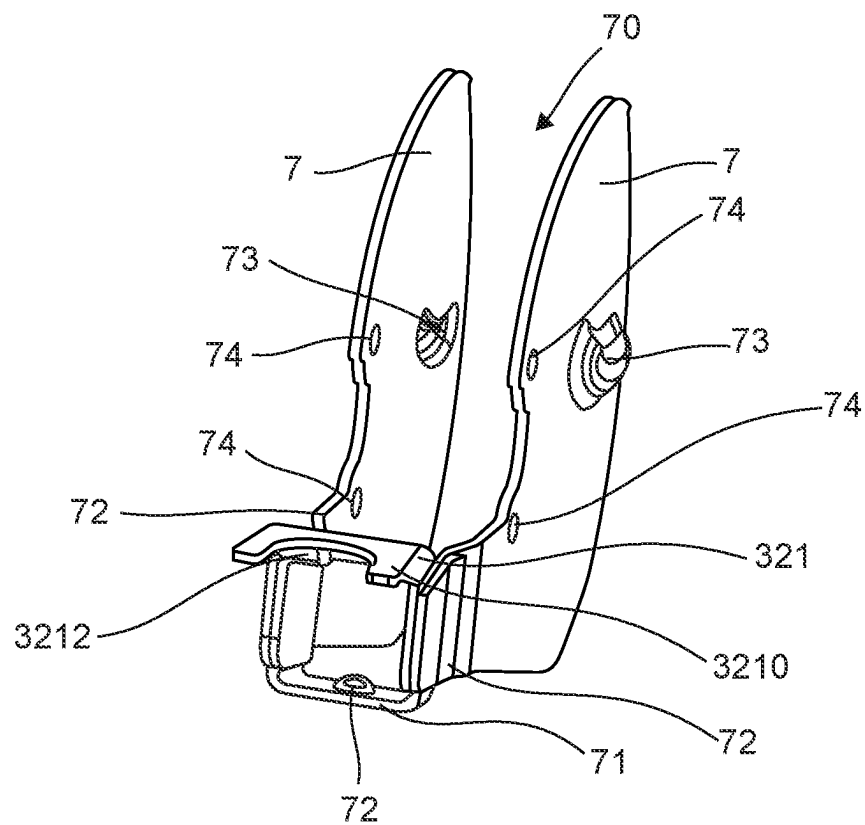


Fig. 8

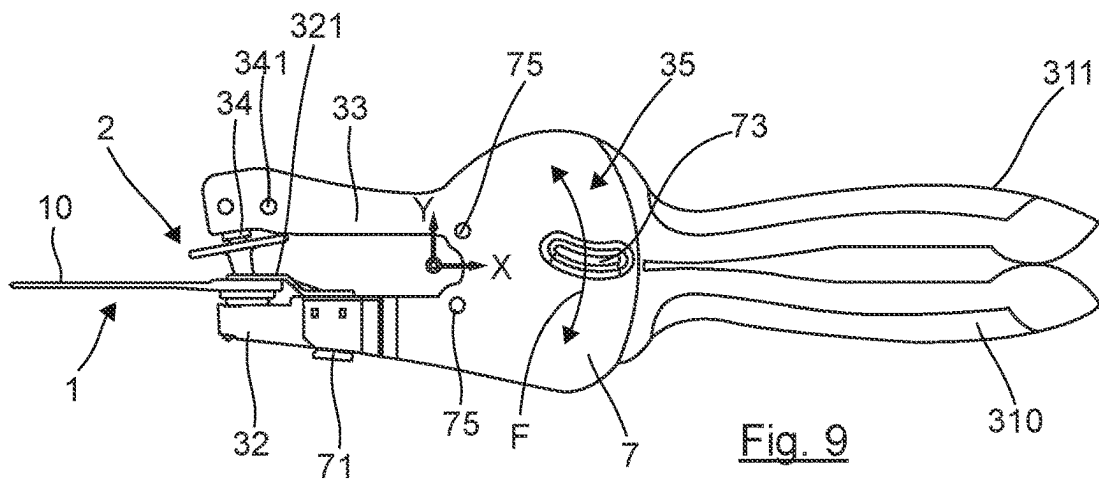


Fig. 9

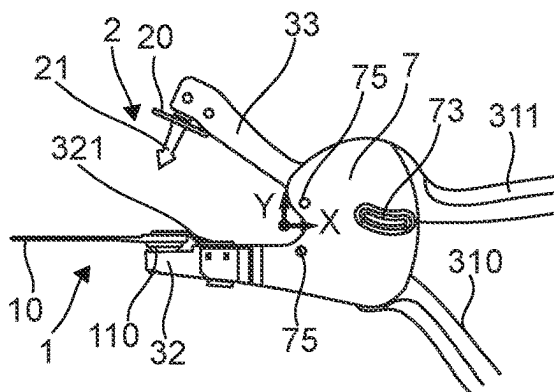


Fig. 11

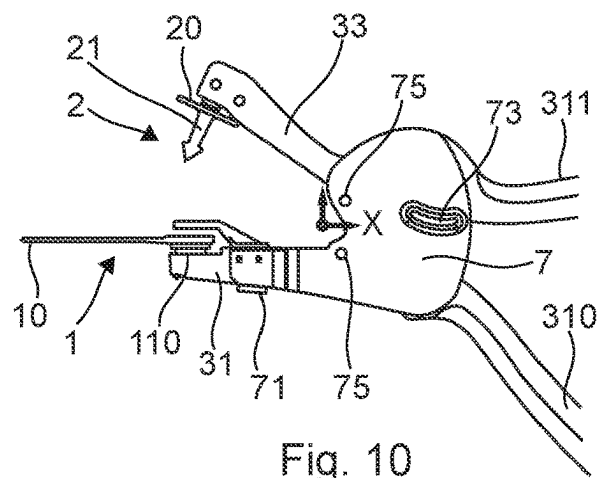


Fig. 10

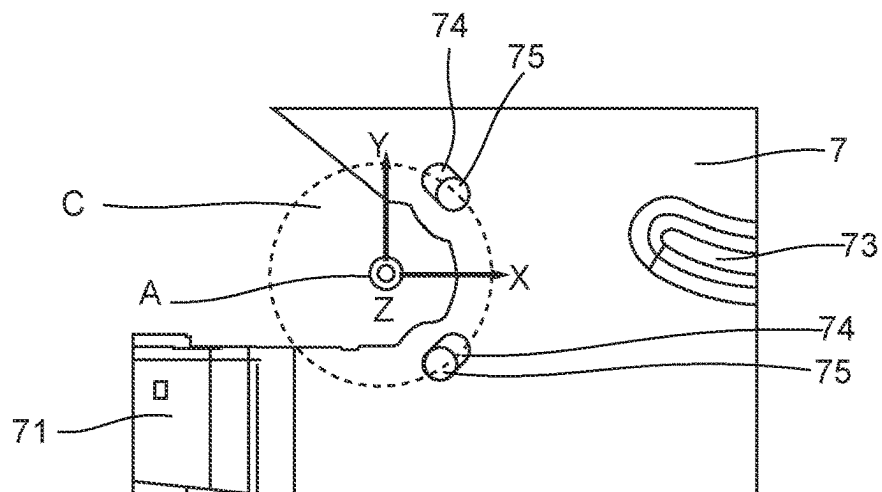


Fig. 12

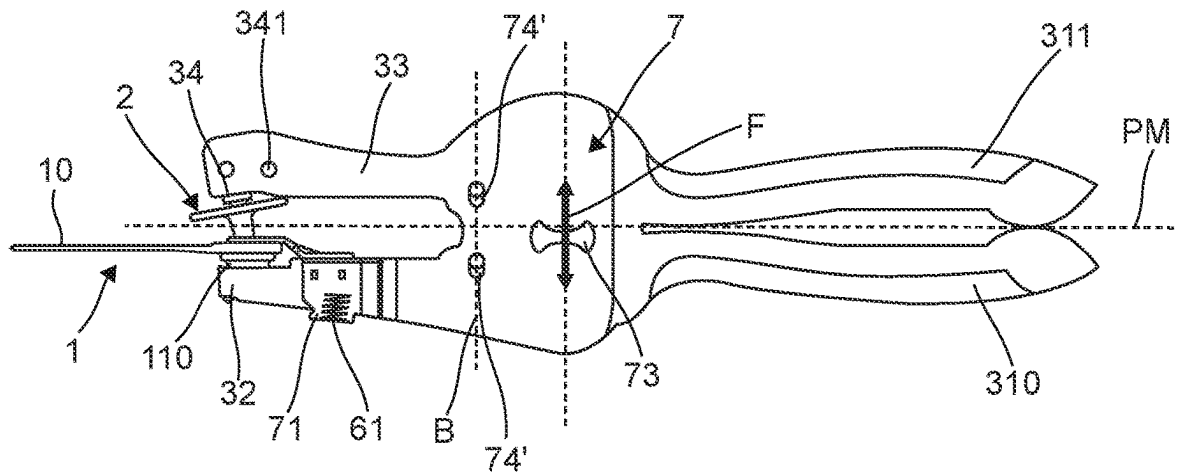


Fig. 13

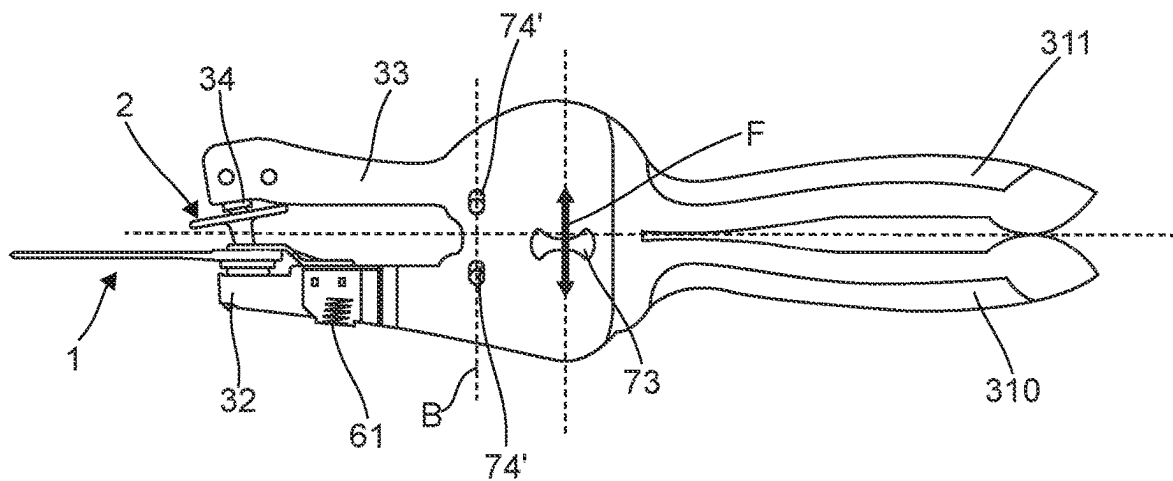


Fig. 14

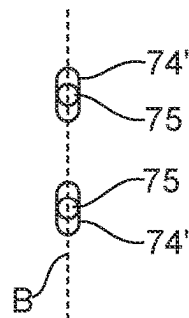


Fig. 15

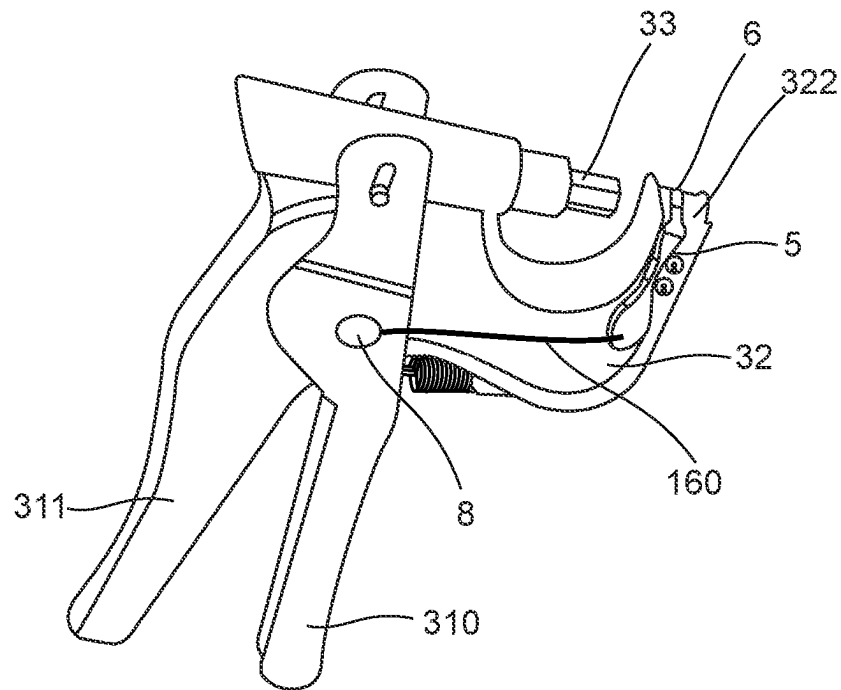


Fig. 16

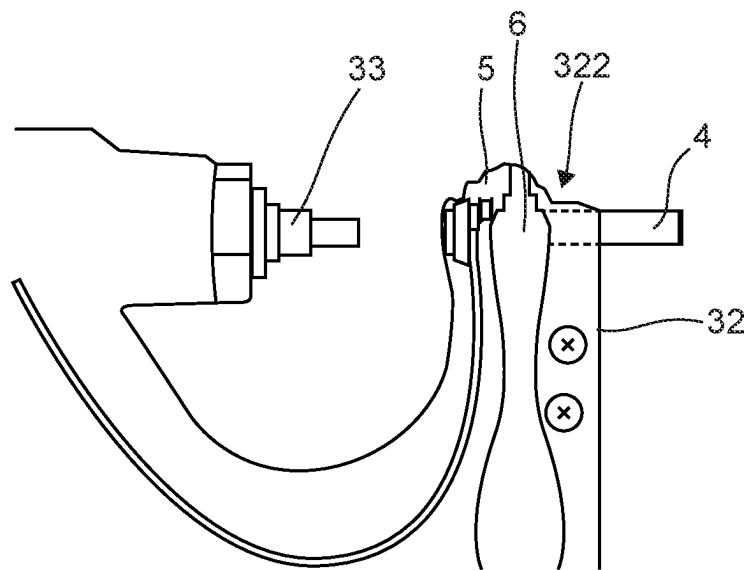


Fig. 17

1

GRIPPER FOR MANIPULATING A DEVICE FOR IDENTIFYING AN ANIMAL AND/OR REMOVING TISSUE FROM AN ANIMAL COMPRISING HOLDING MEANS WITH REMOTE ACTUATION

1. FIELD OF THE INVENTION

The invention relates to the field of design and manufacture of clamps allowing to secure, to the ear of an animal, an identification device bearing visual and/or electronic identification information relating to this animal. It also relates to the field of clamps allowing to handle a device for sampling an animal tissue.

More specifically, the invention relates to the manner for holding secured to the jaws of the clamp the elements that compose such an identification and/or tissue sampling device.

2. PRIOR ART

In order to ensure animal identification in particular to meet sanitary and traceability requirements, identification devices are placed on farm animals in particular pigs, cattle, sheep/goats or others.

Among these identification devices, mention may be made to earrings.

Conventionally, these earrings comprise a female element and a male element intended to be placed on either side of an ear and to cooperate together to hold the earring secured to the ear.

As represented in FIGS. 1*a* and 1*b*, respectively illustrating a front view and a side view of an example of a female element, such an element 1 herein comprises an information carrier plate 10 prolonged at one of its end by a locking housing 11 closed by a cap 110.

As represented in FIGS. 2*a* and 2*b*, respectively illustrating a front view and a side view of an example of a male element, such an element 2 herein comprises a plate 20 prolonged at one of its ends by a locking tip 21 intended to cooperate with the locking housing 11 of the female element 1.

At least one, or both, of the female and male elements may carry information enabling in particular the identification of the animal. For example, this information may consist of visually legible markings (alphanumeric markings, barcode . . .) and/or of information recorded on an electronic medium such as an RFID tag or others.

The set-up of such an earring implies the use of a clamp designed to this end as illustrated in FIG. 3.

A clamp 3 of this type comprises a handle 31 comprising two arms 310, 311 hinged to one another and each prolonged by a jaw 33, 32 respectively.

The jaw 33 carries a punch 34 along which a male element 2 can be forcibly fitted by inserting the punch 34 into a housing with a complementary shape 210 formed inside the locking tip 21.

The jaw 32 comprises a receptacle 320 suitable for housing the cap 110 of a female element 1. It also carries a flange 321 (also called pedal) intended to hold a female element 1 secured to the jaw 32.

This flange 321 comprises a fork-like shaped end 3210 with a front opening enabling the passage of the locking tip 21 of a male element 2. It extends towards the joint area of the arms 310, 311 by an actuation tab 3211 on which an operator can act to bring the fork 3210 away from or close

2

to the end of the jaw 32, a spring tending to bring the fork 3210 close to the end of the jaw 32.

The set-up of such an identification earring on the ear of an animal is achieved in the following way.

5 An operator seizes with his first hand the handle 31, then fits with his second hand a male element 2 along the punch 34. Afterwards, he seizes a female element 1 in his second hand and brings it close to the jaw 32 in order to insert the cap 110 into the receptacle 320. For this purpose, he dis-
10 places his first hand so as to seize the clamp no longer by the handle 31 but at the jaw 32 so as to be able to act on the actuation tab 3211 to raise the fork 3210 with his first hand and thus clear access to the receptacle 320. The operator can then position the female element 1 with his second hand
15 between the jaw 32 and the flange 321 by inserting the cap 110 into the housing 320. Finally, the operator releases the actuation tab 3211 such that the flange 321 gets close to the jaw 32 by the effect of the spring to hold the female element
20 1 secured to the jaw 32.

Afterwards, the operator places an ear 0 between the two jaws 32, 33, then actuates the handle 31 to bring the jaws close to one another until the locking tip 21 fits into the locking housing 11.

25 Thus, the female element 1 and the male element 2 grip the ear 0 such that the earring is irreversibly secured therein, as represented in FIG. 4.

Although effective and relatively simple to implement, this technique may nevertheless be improved even more.

30 Indeed, it is not always obvious for the operator to hold the clamp and handle the flange 321 with one hand and handle the female element 1 with the other hand. For this purpose, in general, the operator seizes the clamp with one hand, not by the handle 31 but in the region of the joint of the two arms 310, 311, so as to be able to hold the clamp and handle the actuation tab 3211 of the flange 321 with the same hand and handle the female element 1 with the other hand. Nonetheless, this practice is not optimal to the extent that the operator does not then hold the clamp comfortably by its handle 31, which has a negative impact on productivity.
40 Hence, there is a need to improve in particular the ergonomics of clamps enabling the set-up of identification earrings. This need also exists for clamps enabling handling a device for sampling an animal tissue, such a device comprising a sampling tube (receptacle) having to be reversibly secured to one of the jaws of the clamp during sampling.

3. OBJECTIVES OF THE INVENTION

50 In particular, it is an objective of the invention to provide an effective solution to at least some of these different problems.

In particular, according to at least one embodiment, it is an objective of the invention to provide a clamp for handling a device for identifying an animal and/or sampling an animal tissue which allows to facilitate the work of an operator performing this task, and in particular to improve ergonomics.

60 In particular, it is an objective of the invention, according to at least one embodiment, to provide such a clamp which enables the operator to comfortably hold the clamp by its handle throughout the set-up of an identification earring on the ear of an animal, and in particular during the set-up of the male and/or female elements on the clamp, or the completion of a tissues sampling.

It is another objective of the invention, according to at least one embodiment, to provide such a clamp that has a

simple and/or robust and/or economical design and/or whose use contributes to improving productivity.

4. DISCLOSURE OF THE INVENTION

For this purpose, the invention provides Clamp for handling a device for identifying an animal and/or sampling animal tissue, said clamp comprising two jaws, a handle for moving said jaws together, and means for securing an element of said identification and/or sampling device to one of said jaws, said securing means being movable between:

a release position wherein they do not secure said element of said identification and/or sampling device to said jaw, and

a securing position wherein they secure said element of said identification and/or sampling device to said jaw.

According to the invention, said clamp comprises actuation means for moving said securing means from one to the other position thereof, said actuation means being actuable via said handle.

Thus, according to this aspect of the invention, an operator can comfortably grip the handle and act on the actuation means with the same hand to achieve securing of one of the elements of an identification earring. Thus, he can set the element of the identification earring in place with the other hand to secure it in the clamp. In the case of a sampling device, the operator can thus comfortably handle the clamp with one hand while he brings the sampling tube close with the other hand to hold it secured to the clamp.

Thus, the ergonomics are improved by enabling the operator to handle the clamp comfortably with one hand and handle the portions of the earring or of the sampling tube with the other hand.

Thus, the technique according to the invention allows go improve comfort of the operator and to improve productivity by helping him save time.

According to a possible variant, said identification device comprises a female element and a male element intended to be placed on either side of an animal's ear and to be moved together to engage with one another and secure said identification device to said ear, said female and male elements being each intended to be supported by one of said jaws, said securing means being movable between:

a release position wherein said securing means are moved away from said corresponding jaw, and

a securing position wherein said securing means are moved towards said corresponding jaw to secure if necessary said male element or said female element between said securing means and said corresponding jaw.

In this case, an element of an identification device may be effectively secured between the securing means and the corresponding jaw.

According to a possible variant, said securing means are movably mounted about a first axis of rotation between the securing position thereof and the release position thereof.

According to a possible variant, said securing means are movably mounted along an axis of translation between the securing position thereof and the release position thereof.

According to a possible variant, said handle comprises two arms rotatably mounted along a second axis of rotation, said first axis of rotation (Z) being essentially parallel with said second axis of rotation or said axis of translation (B) being essentially orthogonal to said second axis of rotation.

In some variants, the axis of rotation of the securing means may be coincident with that of the arms or the axis of translation of the securing means may be secant to the axis

of rotation of the branches. The axis of rotation of the securing means may be coincident with that of the arms or the axis of translation of the securing means may also be remote from the axis of rotation of the arms.

A clamp according to the invention may comprise elastic return means tending to return said securing means to one position thereof, the actuation means being used to move same to the other position thereof.

According to a possible variant, said securing means comprise a flange movable between:

said release position wherein said flange and said corresponding jaw are moved apart to release said male element or said female element, and

a securing position wherein said flange and said corresponding jaw are moved together to grip said male element or said female element between said flange and said corresponding jaw.

According to a possible variant, said actuation means comprise at least one lever extending to the end of said handle situated on the side of said jaws so as to be suitable for being actuated by a finger of a hand grasping said handle.

This allows to improve ergonomics.

According to a possible variant, said at least one lever is connected to a flank to which said flange is connected.

According to a possible variant, said lever forms with said flank and said flange a one-piece assembly.

According to a possible variant, said flank bears at least two slots stretching along a same circle wherein the centre passes via the first axis of rotation of the securing means, said slots housing mounting axis for engaging said flank with the rest of said clamp.

According to a possible variant, said flank bears at least two oblong slots stretching along said axis of translation of the securing means, said oblong slots housing mounting axis for engaging said flank with the rest of said clamp.

According to a possible variant, said slots or said oblong slots are laterally offset from said first axis of rotation towards said handle.

This allows to clear more space between the jaws to accommodate an ear therein.

According to a possible variant, said sampling device comprises a receptacle suitable for receiving a sample, said securing means comprising two grips placed facing one another movably mounted between:

a release position wherein said grips are moved apart to release said receptacle, and

a securing position wherein said grips are moved together to secure said receptacle to said corresponding jaw.

According to at least one embodiment, the securing means comprise at least one securing element (for example a flange or a pair of grips) movable between:

a release position in which it does not hold said element of said identification and/or sampling device secured to said jaw, and

a securing position in which it holds said element of said identification and/or sampling device secured to said jaw by gripping.

According to this embodiment, said clamp comprises actuation means to displace said at least one securing element from one to the other position thereof, said actuation means comprising at least one actuation element (for example a lever or a button) that can be manually actuated from the handle by means of a hand gripping the handle.

5. LIST OF FIGURES

Other features and advantages of the invention will appear on reading the following description of particular embodi-

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ments, provided simply as illustrative and non-limiting examples, and from the appended drawings amongst which:

FIGS. 1a and 1b illustrate front and side views of a female element of an identification earring;

FIGS. 2a and 2b illustrate front and side views of a male element of an identification earring;

FIG. 3 illustrates a clamp according to the prior art for the set-up of an identification earring;

FIG. 4 illustrates securing of an identification earring to the ear of an animal;

FIG. 5 illustrates a side view of a clamp according to the invention, in the closed position, on which an identification earring is set in place, and whose securing means are in the securing position;

FIG. 6 illustrates a partial longitudinal sectional view of the clamp of FIG. 5;

FIGS. 7 and 8 illustrate perspective views of a clamp according to the invention;

FIG. 9 illustrates a side view of a clamp according to the invention in the closed position, in which an identification earring is set in place, with the flange in the release position;

FIG. 10 illustrates a side view of a clamp according to the invention in the open position, on which an identification earring is set in place, with the flange in the release position;

FIG. 11 illustrates a side view of a clamp according to the invention in the open position, on which an identification earring is set in place, with the clamp in the securing position;

FIG. 12 illustrates a detail view of FIGS. 10 and 11;

FIGS. 13 and 14 illustrate side views of a clamp in the closed position with securing means displaceable in translation respectively in the securing position and in the release position;

FIG. 15 illustrates a detail of the oblong slots of the clamp illustrated in FIGS. 13 and 14;

FIGS. 16 and 17 illustrate a variant of a clamp for handling a tissue sampling device.

6. DESCRIPTION OF PARTICULAR EMBODIMENTS

6.1. Architecture of a Clamp

An embodiment of a clamp according to the invention for the set-up of an identification earring on the ear of an animal is presented with reference to FIGS. 5 to 12.

Conventionally, such a clamp 3 comprises a handle 31 comprising two arms 310, 311 each prolonged by a jaw 32, 33.

Conventionally, the two arms 310, 311 are linked by a joint 35 comprising a pivot connection such that bringing the arms 310, 311 close to one another by displacing one of them relative to the other according to the axis of this pivot connection is accompanied by bringing the jaws 32, 33 together, and vice versa.

The jaw 32 carries a punch 34 along which a male element 2 can be forcibly fitted by inserting the punch 34 into a housing with a complementary shape 210 formed inside the locking tip 21.

In this embodiment, the punch 34 is mounted on a support 340 movably mounted in the jaw 33 according to the axis 341 of a pivot connection essentially parallel to the axis of the joint 35 of the arms 310, 311. In one variant, the pivot 34 may be fixed relative to the jaw 33 to which it is secured.

The jaw 32 comprises a receptacle 320 suitable for housing the cap 110 of a female element 1. It also carries a flange 321 (also called pedal) intended to hold a female element 1 secured to the jaw 32.

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This flange 321 comprises a fork-like shaped end 3210 with a front opening 3212.

The flange 320 is secured, at its end opposite to that of the fork 3210, to two flanks 7 essentially parallel and distant from one another to form an internal space 70.

Each of these two flanks 7 is prolonged by an extension 72 at the end of which they are linked by a bridge 71. A boss 72 is formed on the bridge 71.

An actuation lever 73 is formed at the surface of each flank 7. In this instance, more particularly, it consists of an actuation boss. Implementing an actuation boss 73 on each flank enables actuation by a right-handed as well as by a left-handed or actuation by two different fingers (for example the thumb and the index). In another variant, one single flank may be implemented. The surface of the actuation bosses 73 intended to be in contact with a finger may be curved to substantially conform to the shape of the finger and thus improve ergonomics.

The internal space 70 is intended to house the rest of said clamp 1 at the joint 35 of the arms 310, 311.

The two flanks 7, the extensions 72 and the bridge 71 are formed from a stamped and folded sheet metal to which the flange 320 is secured by crimping at the end of the extensions 72 to form a one-piece assembly. A fastening means other than crimping, such as welding, screwing, gluing or others may be implemented to ensure securing of the flange. In one variant, the flange may also be formed from the sheet metal of the flanks and folded.

Each flank 7 is crossed by two slots 74 (it is nonetheless possible to implement only one). These slots 74 are oblong and stretch along a circle C with a centre A. They are essentially shaped like a ring portion (crescent portion). They house axes 75 for securing the flanks 7 to the rest of the clamp 1.

Thus, the flange 320 is mounted movable in rotation about a first axis of rotation 2 passing via the centre A of the circle C and essentially parallel with the axis of the pivot connection 35 of the joint of the arms 310, 311.

In this case, the axes of rotation of the flange 320 and of the arms 310, 311 are different. Nonetheless, they may be coincident.

The oblong slots 74 are laterally offset from the first axis of rotation Z towards the handle 31, which allows clearing more space between the jaws 32, 33 to accommodate an ear therein.

Thus, the flange 321 is movable between:

a release position in which the flange 320 and the jaw 32 are moved apart to release the female element placed therebetween, and

a securing position in which the clamp 320 and the jaw 32 are moved together to hold the female element gripped between the flange and the jaw 32.

A compression spring 61 placed on the boss 73 between the bridge 71 and the jaw 32 tends to return the flange 320 back to its securing position.

An operator can act on the actuation bosses 73 to modify the position of the flange 320.

The actuation bosses 73 are placed such that when an operator grips the handle 31 with one hand, these actuation bosses “fall” beneath a finger of this hand, in particular the thumb or the index. Thus, the actuation bosses 73 can be actuated by one finger of the hand gripping the handle 31.

In one variant, illustrated in FIGS. 13 to 15, the flange 320 is movably mounted between its positioned along an axis of translation B essentially orthogonal to a midplane PM of the

clamp extending between its jaws and its arms when the clamp is closed and passing via the axis of rotation of the joint **35** of the arms **32, 33**.

In this case, the oblong slots **74'** stretch according to this axis of translation **B**.

Of course, the axis of translation may be directed differently (for example it may be inclined with respect to the midplane **PM**, or else extend in this plane—in this case the displacement of the flange from one to the other position thereof may be done by sliding from the right to the left in the figures—, and orthogonal to the axis of the joint of the arms) provided that it will allow to carry out a displacement of the flange from one to the other position thereof. In the case of a sliding in the plane **PM** or in a plane essentially parallel therewith, indicating that in the release position, the securing means are away from the corresponding jaw, and that in the securing position, they are close to the corresponding jaw, means that they are brought respectively close to or away from the end of the jaw.

6.2. Operation of a Clamp

The set-up of an identification earring on the ear of an animal is achieved in the following way.

An operator seizes with his first hand the handle **31**, then fits with his second hand a male element **2** along the punch **34**.

The flange **321** lies in its securing position in which it is secured by the effect of the spring **61**. The flange **320** is then brought close to the jaw **32**.

Afterwards, the operator acts on an actuation boss **73** with a finger of the hand with which he comfortably grips the handle **31** so as to place the flange **321** in its release position in which it is moved apart from the jaw **32** through a rotational or translational movement (cf. the arrows **F** in FIGS. **9** and **13**). Access to the housing **320** is then cleared.

Afterwards, the operator seizes a female element **1** in his second hand, brings it close to the jaw **32** and then positions the female element **1** between the jaw **32** and the flange **321** by inserting the cap **110** into the housing **320** with this hand.

The operator releases the actuation boss **73** such that the flange **321** returns back in its securing position by the effect of the spring **61**. The female element **1** is then gripped between the flange and the jaw **32** and is thus secured in position.

Afterwards, the operator places an ear **0** between the two jaws **32, 33**, then actuates the handle **31** to bring the jaws close to one another until the locking tip **21** fits into the locking housing **11**.

Thus, the female element **1** and the male element **2** grip the ear **0** such that the earring is irreversibly secured therein, as represented in FIG. **4**.

Afterwards, the operator releases the handle **31** to open the clamp and then clear it from the animal.

A system for automatically opening the clamp may also be provided. At the end of the closure stroke of the arms **310, 311**, such a system allows to obtain a reopening of the jaws **32, 33** without opening the arms **310, 311**.

6.3. Variants

In the above-described embodiment, the elastic return means tend to return the flange to its securing position and the actuation means are actuated to place them in their release position. This operation may be reversed.

The implementation of securing means with a remote control has just been described in the context of a clamp for handling a device for identifying an animal. Nonetheless, such securing means with a remote control may also be

implemented in the context of a clamp for handling a device for sampling a tissue of an animal such that illustrated in FIGS. **16** and **17**.

In this case, one of the jaws, in this case the jaw **32**, is fixed whereas the other jaw **33** is movable so as to be brought close to or away from the jaw **32**.

The jaw **32** has an end lateral opening **322** into which a sampling tube **4** (receptacle) can be laterally inserted.

In this instance, the jaw **33** is a piston which allows cutting and pushing a tissue of an animal in the sampling tube **4**.

The sampling tube **4** may be held secured to the jaw **32** by means of a pair of grips **5, 6** placed facing one another on either side of the opening **322**.

These grips **5, 6** are movably mounted between:

a release position in which their free ends are moved apart from one another to clear the opening **322** and thus enable the lateral introduction of a sampling tube **4** into the latter or else extract a sampling tube **4** from the latter;

a securing position in which their free ends are moved together to close the opening **322** and thus keep a sampling tube **4** gripped therein.

An actuation button **8**, shifted at the end of the arms **310, 311** allows to actuate the grips **5, 6** remotely to make them move from one to the other position thereof.

This actuation button **8** is linked to the grips **5, 6** by means of a transmission **160** so that the actuation of the button **8** allows to displace the grips **5, 6** from one to the other position thereof. For example, it may consist of a flexible and/or rigid transmission. For example, it may consist of a transmission with one or more rod(s), cable(s), etc.

To perform a sampling, an operator seizes the clamp with one hand and, while the jaws **32, 33** are away from one another, he acts on the actuation button **8** to place the grips **5, 6** in their release position. Then, he can position with the other hand a sampling tube **4** in the lateral opening **322** and hold it therein and then acts on the actuation button **8** with the other hand to place the grips **5, 6** in their securing position. Afterwards, the operator can place a portion of an animal, such as an ear, between the jaws **32, 33**. Afterwards, he actuates the arms **310, 311** so as to bring the jaws **32, 33** close to one another. The jaw **33** then cuts a piece of the ear and pushes it in the sampling tube **4**. Afterwards, the operator releases the clamp to bring the jaws away from one another and clear the clamp from the animal and then acts on the actuation button **8** to place the grips **5, 6** in their release position and thus recover the sampling tube **4**.

In one variant, the actuation button **8** will enable an operator to place the grips **5, 6** in their securing position against the effect of elastic return means. In this case, releasing the actuation button **8** will enable the grips **5, 6** to return back to their release position by the effect of the elastic return means. This operation may be reversed.

Some sampling devices also enable the placement of an identification device.

The invention claimed is:

1. A Clamp for handling an identification device for identifying an animal, said clamp comprising two jaws; a handle comprising two arms linked by a joint for moving said two jaws together and; securing means for securing an element of said identification device to one of said two jaws; said securing means comprising a flange, at least one flank connected to the flange, said at least one flank extending to the joint, and

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at least one actuation means connected to the at least one flank said securing means being movable between at least two positions:
 a release position wherein said securing means does not secure said element of said identification device to the one of said two jaws, and
 a securing position wherein said securing means holds said element of said identification device secured to the one of said two jaws by gripping,
 characterized in that said actuation means actuates said securing means from one of said at least two positions to the other position, said at least one actuation means being manually actuatable by means of a hand gripping one of the arms of the handle.

2. The Clamp according to claim 1, wherein said securing means are movable between:
 said release position wherein said securing means are moved away from a corresponding jaw, and
 said securing position wherein said securing means are moved towards said corresponding jaw.

3. The Clamp according to claim 2, wherein said securing means comprise the flange movable between:
 said release position wherein said flange and the corresponding jaw are moved apart to release a male element or a female element, and
 said a securing position wherein said flange and said corresponding jaw are moved together to grip said male element or said female element between said flange and said corresponding jaw.

4. The Clamp according to claim 1, wherein said at least one actuation means comprise at least one lever extending to an end of said handle situated on a side of said two jaws so as to be suitable for being actuated by a finger of said hand gripping one of the arms of said handle, said at least one lever is-being connected to the at least one flank to which said flange is connected.

5. The Clamp according to claim 1, wherein said securing means are movably mounted about a first axis of rotation between the securing position thereof and the release position thereof.

6. The Clamp according to claim 5, wherein said two arms of said handle are rotatably mounted along a second axis of

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rotation, said first axis of rotation being essentially parallel with said second axis of rotation.

7. The Clamp according to claim 5, wherein said at least one flank includes at least two slots stretching along a circle wherein a center of said circle lies on the first axis of rotation of the securing means.

8. The Clamp according to claim 7, wherein said slots are laterally offset from said first axis of rotation towards said handle.

9. The Clamp according to claim 1, wherein said securing means are movably mounted along an axis of translation between the securing position thereof and the release position thereof.

10. The Clamp according to claim 5 wherein said at least one flank bears at least two oblong slots stretching along said axis of translation of the securing means.

11. The Clamp according to claim 10, wherein said oblong slots are laterally offset from said first axis of rotation towards said handle.

12. The Clamp according to claim 1, comprising elastic return means configured to return said securing means to one of said at least two positions, the at least one of actuation means configured to move said securing means to the other of said at least two positions.

13. The Clamp according to claim 1, wherein said at least one actuation means comprise at least one lever.

14. The Clamp according to claim 13, wherein said lever is suitable for being actuated by a finger of said hand gripping at least one of the arms of said handle.

15. The Clamp according to claim 1, wherein said securing means is a one-piece assembly.

16. The Clamp according to claim 5, wherein said securing means comprises a one-piece assembly, and said at least one flank includes at least two slots stretching along a circle wherein a center of said circle lies on said first axis of rotation.

17. The Clamp according to claim 16, wherein said slots are laterally offset from said first axis of rotation towards said handle.

18. The Clamp according to claim 1, wherein the at least one actuation means is suitable for being actuated by a finger of said hand gripping at least one of the arms of said handle.

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